

Math 275: Calculus III

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Office Hours: MWF 2:30–3:30pm (or by appt.)

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Class Hours: MWF 1:30–2:20pm

Class Room: TLC 249

Course Description

Math 275 is the third and final part of the standard three-semester calculus sequence. The distinct feature of this part of the course is its focus on the multi-dimensional analyses, as opposed to one-dimensional analyses that you learned in Math 170 (Calculus I) and Math 175 (Calculus II). This semester you will learn important concepts such as a vector, a vector field, a function of several variables, partial derivative, a line-integral and multi-variable integrals. You will see that these concepts—as scary as they may sound—are a natural generalization of the things you have previously learned. The ideas of the vector calculus apply to numerous areas of human knowledge such as engineering, physics, pure mathematics, biology, and many others. Some of the application will be seen in the course, some will surface later in your future courses, and others may not be seen until needed in the “real world.”

The structure of the class is as follows. Class periods will be used for minimal lecturing and strategies for solving problems. You should be prepared to participate in discussions by reading the section *prior to* class; practicing some problems prior to class is also beneficial. Learning assessment is based on *weekly quizzes on Friday*, 3 exams, and a final (comprehensive) exam (see below for details). Dates of the 3 exams will be determined based on progress of the class and announced at least a week in advance (they will roughly occur after every 4 weeks). I will drop your lowest quiz score.

There is no assigned or graded homework for this course. At the bottom of the syllabus you will find a list of suggested problems from most of the sections we will cover. I highly recommend you *complete as many of the problems as needed until you feel you understand the material*. Importantly, the quiz given on Friday will use several of the problems from the homework set, *verbatim*. It is critical that you develop good study habits and practice problems on your own time. Answers are provided to many (if not all) of the odd problems in the back of the book. For any other problems, I recommend using some form of AI or a tutor to get help and answers, if needed.

AI Usage

For better or worse, you now have the world's most powerful tutor at your fingertips via various AI platforms. If used properly, such platforms can potentially help you understand how to solve the complex problems that arise in multi-variable calculus. That being said, I would strongly warn against reading solutions provided by AI and then thinking you yourself understand how to solve such problems. Without actively challenging yourself, you will not learn the material. Obviously, all assessment will be done in class without the aid of any electronics therefore whether you choose to obtain help from AI is of little relevance to your grade (beyond whether it helps or hinders you to achieve your learning goals). Here is a quick example of using AI to solve a problem from chapter 15:

- What is the area of the domain given by $D = \{(x,y) \mid 0 \leq x \leq 2\pi \text{ and } \sin(x) + x \leq y \leq \sin(2x) + 2x + 2\}$?
- Can you give a step-by-step solution to the previous question?

The first question to the prompt (above) will give you a cursory overview of the solution (and the solution) to the problem. The second question will you give you a more detailed view of the logic needed to solve the problem. As I mentioned before, this can provide a valuable aid to your understanding of the materials in this course, but it is also easy to convince yourself you know the material when you have not actually properly solved the problems on your own.

Required Materials

- *Calculus*. 9th edn. by J Stewart, D. Clegg, and S. Watson. Cengage Learning: Boston.

Prerequisites/Corequisites

Prerequisites: Math 175

Topics

1. Vectors and Geometry of Space (Chapter 12)
2. Vector Functions (Chapter 13)
3. Partial Derivatives (Chapter 14)
4. Multiple Integrals (Chapter 15)
5. Vector Calculus (Chapter 16)

Course Objectives/Learning Outcomes

Successful students will:

1. Develop the ability to reason geometrically in more than two dimensions, including mastery of basic notions of vector geometry and equations of lines, planes, and spheres.
2. Be able to think geometrically about the graphs of functions of two variables, including the significance of partial derivatives, directional derivatives, and gradients.
3. Learn to set up and evaluate integrals in two and three variables, including working with any of the standard coordinate systems.
4. Understand and correctly interpret the basic notions of vector calculus, including line integrals of vector fields and Green's Theorem.

Final Exam Date

15 December (Monday), 12:45–2:45pm

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\%$ = A, $(93,90\%]$ = A-, $(90,87\%]$ = B+, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 20% of your grade will be determined by weekly quizzes.
- 60% of your grade will be determined by 3 equally-weighted exams.
- 20% of your grade will be determined by the final exam.

Problem Sets

§12.1: 1-8; 9-10; 11-12; 13-18; 19-22; 27-42; 43-46

§12.2: 1-8; 9-14; 15-18; 19-22; 23-26; 27-28

§12.3: 1-10; 15-20; 21-22; 23-24; 39-44

§12.4: 1-7; 9-12; 13; 17-22; 27-32; 33-34

§12.5: 1-5; 6-12; 19-22; 23-40; 45-47; 69-74

§12.6: 1-2; 3-8; 13-22; 23-30; 33-40

§13.1: 2-3; 3-6; 7-16; 21-24; 25-29; 31-34

§13.2: 3-8; 9-16; 17-20; 21-22; 23-24; 25-28; 37-42

§13.3: 3-8; 15-18; 19-24; 25-27; 31-33; 34-35; 51-52; 53-54

§13.4: 3-8; 9-14; 37-40

§14.1: 1-6; 7-16; 32; 45-52; 61-66; 67-70

§14.2: 5-12; 13-18; 19-30; 31-34; 41-50

§14.3: 4-5; 7-8; 9-36; 37-40; 41-44; 45-46; 47-52; 55-56; 57-64

§14.4: 3-10; 15-22; 23-24; 25, 27

§14.5: 1-2; 3-8; 9-10; 11-16; 21-24; 25-30; 31-34; 35-38

§14.6: 4-7; 8-12; 13-19; 21-25; 27-32; 47-52

§14.7: 1-4; 5-22; 33-40

§14.8: 3-16; 23-26; 27-29; 30-33

§15.1: 1-6; 9-11; 13-14; 15-26; 27-34

§15.2: 1-6; 7-10; 11-14; 17-20; 19-22; 23-28; 55-60; 67-68; 69-70

§15.3: 1-6; 9-16; 17-22; 25-28; 29-37

§15.9: 1-6; 7-10; 11-16; 17-22; 25-30

§16.1: 1-12; 13-18; 19-22; 25-28; 29-30; 31-34

§16.2: 1-8; 9-18; 21-24

§16.3: 1-2; 3-10; 11; 12; 13; 14-15; 17-24; 37-40

§16.4: 1-4; 5-12; 13-18